

R Krishnamurthi

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Education

Indian Institute of Information Technology Design and Manufacturing Kurnool
B.Tech in Computer Science

Kurnool, India
Nov 2022 – Apr 2026

Technical Skills

Languages: Python, SQL

AI/ML: PyTorch, TensorFlow (Basics), Hugging Face, Scikit-learn

Generative AI: LLMs, RAG, Prompt Engineering, LLM Fine-tuning, LangChain, Ollama

Vector Search: FAISS, Embeddings, Hybrid Search

Backend & APIs: REST APIs, FastAPI, Docker

Data Engineering: ETL Pipelines, Feature Engineering, Kafka

Databases: PostgreSQL, MySQL

Evaluation: Ragas, Recall@K, NDCG, Precision, Latency

Experience

Data Science Intern

Meraqui Ventures Pvt. Ltd. (karam.ai)

Apr 2025 – Jul 2025

Pune, India

Internship Certificate

- Integrated AI-driven platform with ERP systems via REST APIs, reducing manual data entry by 15%
- Built scalable ETL pipelines with schema validation, reducing data inconsistencies by 12–15%
- Improved candidate-matching system using feature engineering and LLM fine-tuning, increasing precision by 15%
- Designed API adapters, validation pipelines, and test suites to support reliable AI system deployment

Development Engineering Intern

Treacle Technologies (Incubated at IIT Kanpur)

Jun 2024 – Aug 2024

Remote

Internship Certificate

- Engineered real-time data pipelines processing 100k+ logs/day with custom evaluation metrics
- Built monitoring dashboards using Grafana and Elasticsearch, reducing incident detection time by 40%
- Automated preprocessing pipelines across distributed systems, reducing manual effort by 80%
- Deployed AI workflows using containerized environments (Docker) for reproducibility

Projects

Active Conformal Control: Hardware-Aware Safety for Quantized Vision-Language Models 2026

- Proposed a real-time framework for hallucination mitigation in quantized VLMs using density-chasm detection
- Developed a non-parametric density ratio estimator (ppDRE) for latent drift detection
- Designed a hardware-aware cascade using GPU student and CPU-based teacher (Intel AMX)
- Established conformal prediction thresholds for reliability-aware inference

Enterprise RAG System with Comprehensive Evaluation

Feb 2025

Tech: Python, LangChain, FAISS, Streamlit, Ollama, Ragas, Docker

- Developed a production-grade Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, and Ollama.
- Designed hybrid retrieval with optimized chunking and embedding-based semantic search.
- Evaluated retrieval quality using Ragas, Recall@K, NDCG, Precision, and latency metrics.
- Applied prompt engineering techniques to reduce hallucinations and improve response grounding.
- Containerized the complete system using Docker with REST API integration.

Certifications & Achievements

- **Oracle Cloud Infrastructure 2024 Gen AI Certified** – Score: 92.5% (*Certificate*)
- **2nd Place** – GDG Hackathon (AI/ML Track) - Mar 2025 (*Certificate of Appreciation*)
- **2nd Place** – DataWorks Internal Hackathon (LLM Applications) - Apr 2025 (*Certificate of Appreciation*)